A member of the family is covered by the coset determined by its own certificate

Gilad Chase, Neta Dafni, Yuval Filmus, Nathan Lindzey

arxiv:2210.00245v2; Creative Commons Attribution 4.0 International

Editorial note. The counted statement is the author's labelled environment at source0.tex lines 568-572 and its proof environment opens at line 573; the pairing is by adjacency and the retained proof range is the whole environment.

Editorial note. Preamble: the author's own package lines, theorem-environment family and macro block (source lines 5-29).

Editorial note. Bibliography: the article ships its own compiled bibliography file (main.bbl, a complete thebibliography with 29 entries) and the retained proofs cite keys from it, so the card bundles that file unchanged; no bibliographic entry is edited or re-rendered.

Editorial note. Retained-licence form: this source's licence evidence carries no acceptance record and none may be written on anyone's behalf, so the card points at a task-directory evidence copy in the team-wide four-key reuse form whose basis states that the source's staged record is reused as a material basis only. No status key and no acceptance record are written.

Editorial note. Metadata provenance: source_version is the value recorded in the source's licence evidence (2210.00245v2); the author list is transcribed from the authors_index key of the source record of the pinned copy, and the title from its title key.

Editorial note. The source carries no label-number dump and no counter values are imposed, so the printed numbering is the wrapper's default; the identity of the counted item is its author label.

Editorial note. r002 note: r001 failed with an undefined reference raised from inside the counted proof; r002 adds exactly the environment that the citation names, as a context span with its provenance, and changes nothing else. The article's own compiled bibliography (main.bbl) was already bundled and is unrelated to this failure.

Editorial scope. One original result of the pinned author manuscript, counted once: members of the family are covered by the coset of a minimal certificate (author label lem:covering-sym).

Allowed standard prerequisites (editorial).

- m-covered family. The family is m-covered when all its members are contained in a single coset determined by prescribing m images.. It is the conclusion of the statement.
- Minimal certificate. A certificate of the member that is minimal for the given function, so its size is the value of the certificate-size function at that member.. It is the coset whose size is estimated.
- Intersection bound for representatives. Two members with equal characteristic value admit representatives meeting exactly in the intersection of their certificates.. It is the estimate that forces the second member into the coset.

Lemma 0.1. *Let $f\colon S_n \to \{0,1\}$ be the characteristic function of a 2-intersecting family, and suppose that $C(f) \leq n-2$. If $f(\alpha) = f(\beta) = 1$ and C_α, C_β are minimum certificates for α, β (respectively), then $|C_\alpha \cap C_\beta| \geq 2$.*

Lemma 0.2. *Let f be the characteristic function of a 2-intersecting family $\mathcal{F} \subseteq S_n$, and suppose that $C(f) \leq n-2$.*

If $\alpha \in \mathcal{F}$ then $\mathcal{F}$ is m-covered for $m = C(f,\alpha) - 1$.

Proof. Let $C_\alpha = \{(i_1, j_1), \ldots, (i_{m+1}, j_{m+1})\}$ be a minimal certificate for α. We will show that $\mathcal{F} \subseteq [i_1 \mapsto j_1] \cup \cdots \cup [i_m \mapsto j_m]$.

Suppose that $\beta \in \mathcal{F}$, and let C_β be a minimal certificate for β. According to Theorem 0.1, $|C_\alpha \cap C_\beta| \geq 2$. Therefore $(i_s, j_s) \in C_\beta$ for some $s \in [m]$, implying that $\beta \in [i_s \mapsto j_s] \subseteq [i_1 \mapsto j_1] \cup \cdots \cup [i_m \mapsto j_m]$. □

References

[AK97] Rudolf Ahlswede and Levon H. Khachatrian. The complete intersection theorem for systems of finite sets. *European J. Combin.*, 18(2):125–136, 1997.

[AK99] Rudolf Ahlswede and Levon H. Khachatrian. A pushing-pulling method: new proofs of intersection theorems. *Combinatorica*, 19(1):1–15, 1999.

[BdW02] Harry Buhrman and Ronald de Wolf. Complexity measures and decision tree complexity: a survey. *Theoret. Comput. Sci.*, 288(1):21–43, 2002. Complexity and logic (Vienna, 1998).

[BMRR21] Angelot Behajaina, Roghayeh Maleki, Aina Toky Rasoamanana, and A. Sarobidy Razafimahatratra. 3-setwise intersecting families of the symmetric group. *Discrete Math.*, 344(8):Paper No. 112467, 15, 2021.

[CK03] Peter J. Cameron and C. Y. Ku. Intersecting families of permutations. *European J. Combin.*, 24(7):881–890, 2003.

[CSST08] Tullio Ceccherini-Silberstein, Fabio Scarabotti, and Filippo Tolli. *Harmonic analysis on finite groups*, volume 108 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2008. Representation theory, Gelfand pairs and Markov chains.

[Daf22] Neta Dafni. Complexity measures on the symmetric group and beyond. Master's thesis, Technion — Israel Institute of Technology, 2022.

[DFL+21] Neta Dafni, Yuval Filmus, Noam Lifshitz, Nathan Lindzey, and Marc Vinyals. Complexity measures on the symmetric group and beyond. In *12th Innovations in Theoretical Computer Science Conference*, volume 185 of *LIPIcs. Leibniz Int. Proc. Inform.*, pages Art. No. 87, 5. Schloss Dagstuhl. Leibniz-Zent. Inform., Wadern, 2021.

[EFP11] David Ellis, Ehud Friedgut, and Haran Pilpel. Intersecting families of permutations. *J. Amer. Math. Soc.*, 24(3):649–682, 2011.

[EKR61] P. Erdős, Chao Ko, and R. Rado. Intersection theorems for systems of finite sets. *Quart. J. Math. Oxford Ser. (2)*, 12:313–320, 1961.

[Ell12a] David Ellis. A proof of the Cameron-Ku conjecture. *J. Lond. Math. Soc. (2)*, 85(1):165–190, 2012.

[Ell12b] David Ellis. Setwise intersecting families of permutations. *J. Combin. Theory Ser. A*, 119(4):825–849, 2012.

[FD77] Péter Frankl and Mikhail Deza. On the maximum number of permutations with given maximal or minimal distance. *J. Combinatorial Theory Ser. A*, 22(3):352–360, 1977.

[FHKL16] Yuval Filmus, Hamed Hatami, Nathan Keller, and Noam Lifshitz. On the sum of the L_1 influences of bounded functions. *Israel J. Math.*, 214(1):167–192, 2016.

[Fil17] Yuval Filmus. A comment on intersecting families of permutations. arXiv:1706.10146, 2017.

[FMS21] Shaun Fallat, Karen Meagher, and Mahsa N. Shirazi. The Erdős-Ko-Rado theorem for 2-intersecting families of perfect matchings. *Algebr. Comb.*, 4(4):575–598, 2021.

[GM09] Chris Godsil and Karen Meagher. A new proof of the Erdős-Ko-Rado theorem for intersecting families of permutations. *European J. Combin.*, 30(2):404–414, 2009.

[GM16] Chris Godsil and Karen Meagher. *Erdős-Ko-Rado theorems: algebraic approaches*, volume 149 of *Cambridge Studies in Advanced Mathematics*. Cambridge University Press, Cambridge, 2016.

[GM17] Chris Godsil and Karen Meagher. An algebraic proof of the Erdős-Ko-Rado theorem for intersecting families of perfect matchings. *Ars Math. Contemp.*, 12(2):205–217, 2017.

[Lin17] Nathan Lindzey. Erdős-Ko-Rado for perfect matchings. *European J. Combin.*, 65:130–142, 2017.

[Lin18a] Nathan Lindzey. Intersecting families of permutations of perfect matchints. arXiv:1811.06160, 2018.

[Lin18b] Nathan Lindzey. *Matchings and Representation Theory*. PhD thesis, University of Waterloo, 2018.

[LM04] Benoit Larose and Claudia Malvenuto. Stable sets of maximal size in Kneser-type graphs. *European J. Combin.*, 25(5):657–673, 2004.

[MM05] Karen Meagher and Lucia Moura. Erdős-Ko-Rado theorems for uniform set-partition systems. *Electron. J. Combin.*, 12:Research Paper 40, 12, 2005.

[MR21] Karen Meagher and A. Sarobidy Razafimahatratra. The Erdős-Ko-Rado theorem for 2-pointwise and 2-setwise intersecting permutations. *Electron. J. Combin.*, 28(4):Paper No. 4.10, 21, 2021.

[NÖ03] Sampo Niskanen and Patric R. J. Östergård. Cliquer user's guide, version 1.0. Technical Report T48, Communications Laboratory, Helsinki University of Technology, Espoo, Finland, 2003.

[O'D14] Ryan O'Donnell. *Analysis of Boolean functions*. Cambridge University Press, New York, 2014.

[Sag01] Bruce E. Sagan. *The symmetric group*, volume 203 of *Graduate Texts in Mathematics*. Springer-Verlag, New York, second edition, 2001. Representations, combinatorial algorithms, and symmetric functions.

[Sag22] Sage Developers. *SageMath, the Sage Mathematics Software System (Version 9.5)*, 2022. `https://www.sagemath.org`.